

ImdexTM

Isosorbide-5-mononitrate
60 mg CR Tablet

Qualitative and Quantitative Composition

One controlled release tablet contains:
Isosorbide mononitrate 60 mg

Pharmaceutical Form

Controlled release tablet

60 mg: Yellow, oval, biconvex, film coated, scored on one side and marked with IM on the other side.

Therapeutic indications

Prophylactic treatment of angina pectoris.

Posology and method of administration

Imdex is intended for prophylactic treatment. The dosage is individual and must be titrated according to clinical response. **Imdex** prolonged release tablets are taken once daily, in the morning.

When initiating treatment, in order to avoid headache, 30 mg be given for the first 2-4 days.

Normal dosage: 60 mg daily. If necessary the dose may be increased to 120mg daily. There is a risk of tolerance development with nitrate therapy. It is therefore important that **Imdex** is taken once daily in order to obtain intervals with low nitrate concentrations and thus reduce the risk of tolerance development.

If necessary, **Imdex** may be combined with beta-adrenoreceptor blockers and calcium antagonists.

Administration

The tablet is dividable. The whole or divided tablet of **Imdex** 60mg should not be chewed or crushed and should be swallowed together with ½ glass of fluid.

Contraindications

Shock, hypotension, constrictive cardiomyopathy, constrictive pericarditis and pericardial tamponade.

Hypersensitivity to the active ingredients of any excipients. Concomitant treatment with Phosphodiesterase type 5 Inhibitors (eg sildenafil).

Special warnings and special precautions for use

Caution should be exercised in the treatment of patients with serious cerebrovascular disease, increased intracranial pressure, aortic stenosis, mitral stenosis and hypertrophic obstructive cardiomyopathy, anaemia, hypoxaemia and hypothyroidism.

Interaction with other medicinal products and other forms of interaction

*The following combination with **Imdex** must be avoided:*

Phosphodiesterase type 5 Inhibitors (eg sildenafil): Administration of Phosphodiesterase type 5 Inhibitors (eg sildenafil) is contraindicated to existing treatment with nitroglycerin products since it may lead to serious fall in blood pressure, ischaemia and circulatory disorders with permanent heart and brain damages.

Use during pregnancy and lactation

Pregnancy
Clinical experience from pregnant women limited. Experimental animal data do not indicate an increased risk of foetal damage.

Lactation

No information is available regarding passage into breast milk.

Effects on ability to drive and use machines

Since dizziness may be experienced in **Imdex** treatment this should be considered when strict attention is required e.g. when driving cars or operating machines.

Undesirable effects

The majority of side effects are pharmacologically mediated and dose dependent. Headache may occur in about 25% of the patients when treatment is initiated. This side effect is due to the vasodilatory effect of the product but usually disappears within a week. The headache can be avoided when 30 mg is given the first 2-4 days.

A fall in blood pressure may lead to reflexory tachycardia, dizziness and fainting.

Frequency/ Organ	Very common (>1/10)	Common (>1/100, <1/10)	Less common (>1/1000, <1/100)	Rare (>1/10000, <1/1000)	Very rare (<1/10000)
General		headache, dizziness		fainting	
Circulation		hypotension, tachycardia			
Gastrointestinal		nausea	vomiting, diarrhea		
Skin				rash, pruritus	
Musculoskeletal					myalgia

148mm (w) x 210mm (h)



Black 100%

Overdose

Limited experience. 20 mg given to a 2-year old and 5-year old did not cause any symptoms.

Symptoms: Pulsing headache. Excitation, flushing, cold perspiration, vomiting, dizziness syncope, tachycardia, palpitations and fall in blood pressure. Very large doses may cause methaemoglobinemia (very rare).

Treatment: If necessary gastric lavage, charcoal. If fall in blood pressure occurs intravenous fluid should be given at first hand. (Methylthionin 1-2 mg/kg should be given slowly intravenously in cyanosis as a consequence of methaemoglobinemia). Symptomatic treatment.

Pharmacological Properties**Pharmacodynamic properties**

Pharmacotherapeutic group: Vasodilators in heart diseases, organic nitrates

Imdex is a controlled release dosage form of isosorbide-5-mononitrate, which is an active metabolite of isosorbide dinitrate. Nitrate compounds produce a dose-dependent relaxation of smooth muscle. The effect of treatment is dependent on the dose and individual sensitivity. Low doses produce dilatation of the veins and reduced venous reflux to the heart (reduced preload). High doses in addition cause arterial dilatation and reduced vascular resistance (reduced afterload). Isosorbide-5-mononitrate reduces the work of the heart as a result of venous and arterial dilatation and may also have a direct dilating effect on coronary vessels. By reducing end diastolic pressure and volume, the preparation lowers the intramural pressure, which leads to an improvement of subendocardial perfusion. The net effect of isosorbide-5-mononitrate is reduced heart work and improved myocardial oxygen saturation.

Imdex is intended for prophylactic treatment of angina pectoris. The duration of effect, measured with exercise tests, is at least 12 hours. At this point in time the plasma concentration is at the same level as 1-2 hours after tablet intake (about 1300 nmol/litre).

In continuous treatment with nitrate compounds there is a risk of tolerance development, which varies from one individual to another. **Imdex** must therefore be administered once daily to obtain an interval with low nitrate concentration.

Imdex consists of an insoluble matrix, which usually disintegrates as a result of/through intestinal peristalsis. The tablet may therefore appear intact, whereas the active substance has dissolved during passage through the gastrointestinal tract.

Pharmacokinetic properties

The onset of effect occurs within an hour. The bioavailability of **Imdex** is about 90 %. Absorption is not affected by concomitant food intake. The release of active substance from **Imdex** is gradual and independent of pH, and is completed after about 10 hours. After repeated oral administration of 60 mg once daily, maximum plasma concentration (about 3000 nmol/litre) is achieved after about 4 hours. The plasma concentration then decreases, to be less than 500 nmol/litre at the end of the dosing interval (24 hours after ingestion). The volume of distribution of isosorbide-5-mononitrate is about 0.6 litres/kg and clearance is 115ml/minute. Elimination takes place mainly by denitration and conjugation in the liver to inactive metabolites. The metabolites are excreted mainly via the kidneys. Only about 2 % of a given dose is excreted intact via the kidneys.

Impaired hepatic or renal function have no influence on the clinical effect.

Shelf-life

Please refer to outer package

Special precautions for storage

Store below 30°C in a cool dry place. Protect from light.

Keep out of reach of children

Jauhkan dari kanak-kanak

Route of administration: Oral**Nature and contents of container**

Blister pack:

Controlled release tablets 60 mg

(28 tablets)

Product Registration Holder :**Duopharma Marketing Sdn. Bhd.**

Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13,

Bangi Industrial Estate, 43650 Bandar Baru Bangi,

Selangor, Malaysia.

Manufacturer :**Duopharma Manufacturing (Bangi) Sdn Bhd.**

Lot No. 2 & 4, Jalan P/7, Section 13, Bangi Industrial Estate,

43650 Bandar Baru Bangi, Selangor, Malaysia.

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